

Quantum Mechanics III

Lecture 3

Quantum Channels and Kraus Representations

- Operational setting
- Mathematical representation
- Dynamics and measurement
- Consistency conditions
- Example

■ Operational setting

Preparation and measurement procedures are distinguished from their matrix representations.

$$\mathcal{E}(\rho) = \sum_a K_a \rho K_a^\dagger$$

■ Mathematical representation

Positivity, trace, and tensor-product conventions are fixed before the main calculation.

$$\sum_a K_a^\dagger K_a = I$$

■ Dynamics and measurement

The map from a state to observed probabilities is written explicitly for the experiment under discussion.

$$J_{\mathcal{E}} = (I \otimes \mathcal{E})(|\Phi\rangle\langle\Phi|)$$

■ Consistency conditions

Complete positivity, normalization, Hermiticity, and appropriate classical limits are checked as needed.

■ Example

The example is worked for a qubit or two-mode system before extending the notation to higher dimension.

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Reading: Park, ch. 3; Nielsen–Chuang §8.2; QDC process profile